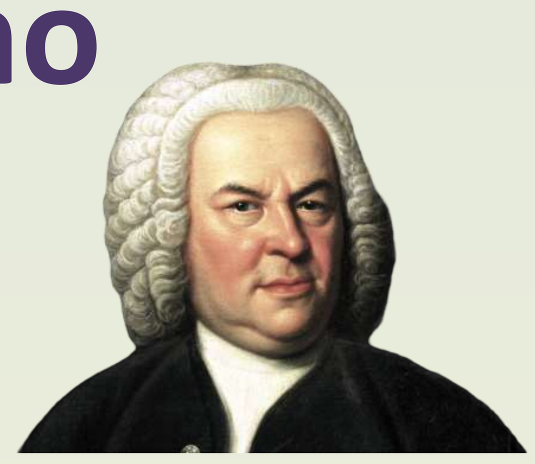
Roman Analysis on Bach's Chorales with no note information

Charalampos Sarakatsanis



I. Introduction

Have you ever thought that behind a piece of music there are harmonic dependencies with relations potentially stronger than the sound itself? This poster presents an analysis of Bach's chorales that models the **harmonic** relations between Roman numerals and their properties (major, minor, inverted, and so on) **without using notes or MIDI data at all**.

Why Roman Numerals?

The harmonic conventions of the Baroque exist to let a composer direct expectation through **perceptually grounded patterns**, expressed as musical rules:

- V → I: resolution, closure
- V → vi: deceptive, suspended
- IV → V: preparation, curiosity

II. Data

- **368** four-voice files found
- **362** chorales used for training
- **29,795** chords

Bar numbers are not unique: in .mxl files the number may stay the same while the suffix changes, so offset is used to traverse a chorale.

III. Methods

The data used to produce the final model was:

Variable	Diff Values	Description
roman_scale_degree	14	Degree with mode (I–VII, i–vii) — prediction target
chord_quality	5	major, minor, diminished, augmented, other
inversion	6	0 = root position, 1–5 = inversions
Has7	2	seventh present or not
Duration	12	0.125 – 8.0 quarter notes
beat_strength	continuous	metrical weight, 1.0 (downbeat) – 0.0625
steps_to_end	integer	chords remaining until the end of the chorale

I. Why 14 degrees, not ~300 figures: ~300 full chord figures → 48% appear < 5 times → 14 degrees × 5 predictors, nothing left out

II. Task — chained, per-target control: Chord n → 5 predictors → context stacked → chord n+1

III. Split — grouped by chorale: 262 train | 50 val | 50 test

IV. Models

Models tested: Markov → Linear → GRU → LSTM → **Chained**

Markov: the probabilities of switching from one Roman numeral to the next give cross-entropy ≈ 2.20 and perplexity ≈ 9 (ln(9) ≈ 2.2) — the transition matrix is on GitHub.

Class imbalance. Degrees are imbalanced: V occurs 581 times in the test set against 132 for VI, roughly 4:1 — a property of the Baroque style, not a sampling artefact. Prior correction (tau = 0.60) gave marginal gains only, so uncorrected predictions are reported.

steps_to_end was not predicted, because practically the time a chorale lasts is affected by parameters outside of the model's reach (time limits for a concert, the length of a scene in an oratorio, etc.). For these reasons, this parameter should be given prior to the model's creation.

Linear — logistic regression on the previous Roman chords: the same information Markov uses, but fitted.

GRU / LSTM — a recurrent network over a window of previous chords, with embeddings for every categorical variable. A huge improvement on the earlier models; GRU and LSTM differ very little, so no deeper architecture was needed.

Chained LSTM — the best model: an LSTM predicting the Roman numeral with all chord properties. After each chord the outputs of all 5 predictors become the next observation, window = 8 — the first 8 chords are read before prediction continues.

V. Results

The best performance for each model was:

Model	CE	PP	top-1	top-3	F1
Random	2.64	14	0.071	0.214	—
Majority class	—	—	0.163	—	—
Markov (1st order)	2.2	9	—	—	—
Linear	2.12	8.4	0.315	0.590	0.261
GRU	1.90	6.7	0.405	0.665	0.336
LSTM	1.89	6.6	0.408	0.669	0.340
Chained LSTM	1.70	5.48	0.483	0.730	0.408

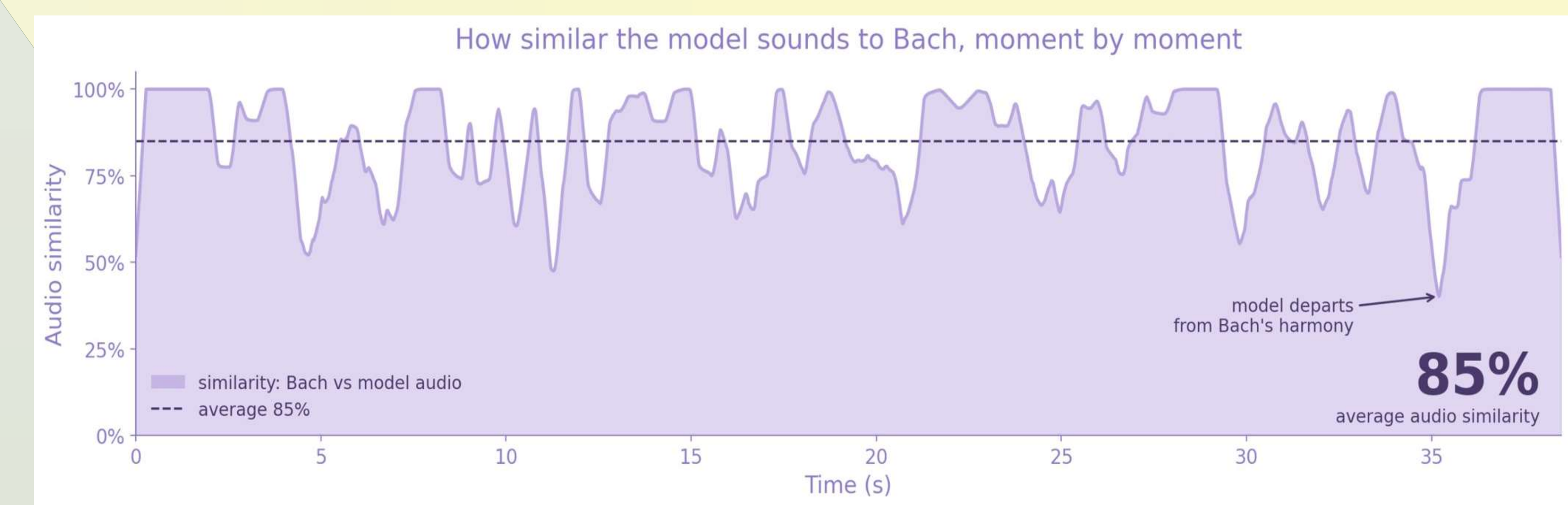
Hyperparameters were fixed rather than searched: window = 8, 128 LSTM units, loss weights (1.0, 0.5, 0.5, 0.3, 0.5). Same seed for everything.

Out of roughly 300 possible complete chord specifications, the final model reproduced Bach's own choice exactly, degree, quality, inversion, seventh and duration all matching the Riemenschneider harmonisation, in 18.3% of cases. The equation of the final model that produces this result is:

$$\mathbf{h}_t = \mathrm{LSTM}(E(x_r),\, E(x_q),\, E(x_i),\, E(x_7),\, E(x_d),\, x_b,\, x_s)_{t-w\,:\,t-1}$$

$$\hat{y}_k = \sigma(W_k\,[\mathbf{h}_t;\; b_t;\; \hat{y}_{<k}]), \qquad k:\; d \to r \to q \to i \to 7$$

Target	Classes	Top-1	Top-3	Macro-F1	Balanced acc.
Roman degree	14	0.483	0.730	0.408	0.405
Chord quality	5	0.624	0.943	0.316	0.307
Inversion	6	0.572	0.905	0.250	0.246
Seventh	2	0.748	—	0.638	0.627
Duration	12	0.744	0.980	0.303	0.275



Chorale BWV 389 (128 chords, the audio demo): 47% of predicted degrees match the original — the first 8 are the shared opening. Translated back into notes the two versions sound very close, so reproducing a piece as notes is far easier than as degrees (intention representation).

VI. Limitations and Future Work

A short analysis was conducted and chorales sharing the same BWV ID were spotted; these existed in two different formats (.krn and .mxl), both following the same **principles** apart from the way they handle repeated bars, so only one copy of each was kept. Two chorales were also found that **were two different harmonic interpretations of the same hymn**

For the future, a great challenge would be using the chorales scale for each chorale to scale all variables in certain ways, in order to avoid further class imbalances and training separate models for major and minor modes would remove the need for the model to infer mode from context.

VII. Conclusions

The emphasis falls on what the composer **intends** the listener to feel — and a story waits behind that intention. The project therefore shifted from reproducing Bach to **continuing his intentions**, and to testing how well a model can learn those moves well enough to generate them.

VIII. References

Cuthbert & Ariza (2010). music21. ISMIR.

Boulanger-Lewandowski, Bengio & Vincent (2012). Modeling temporal dependencies in high-dimensional sequences. ICML.

Hadjeres, Pachet & Nielsen (2017). DeepBach. ICML.

University College Dublin (UCD)
Charalampos Sarakatsanis
Student number – 25267220
18 August 2026

Listen! — Bach vs the model

Visit my Repo